

Numerical Analysis chapter 6 theoretical homework

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I

a

By Simpson's rule, we have $\int_a^b f(x) dx = \frac{b-a}{6}[f(a) + 4f(\frac{b+a}{2}) + f(b)] + R_s$.

x	$f(x)$	$f[x, x+1]$	$f[x, x+1, x+2]$	$f[x, x+1, x+2, x+3]$
-1	$y(-1)$			
0	$y(0)$	$y(0) - y(-1)$		
0	$y(0)$	$y'(0)$	$y'(0) - y(0) + y(-1)$	
1	$y(1)$	$y(1) - y(0)$	$y(1) - y(0) - y'(0)$	$\frac{1}{2}(y(1) - 2y'(0) - y(-1))$

So,

$$\begin{aligned} p_3(x) &= L_0 + L_1(x+1) + L_2x(x+1) + L_3x^2(x+1) \\ &= L_3x^3 + (L_2 + L_3)x^2 + (L_1 + L_2)x + L_0 + L_1 \\ \int_{-1}^1 p_3(x) dx &= \frac{2}{3}(L_3 + L_2) + 2(L_0 + L_1) \\ &= \frac{1}{3}(y(-1) + 4y(0) + y(1)) \end{aligned}$$

It is the same as Simpson's rule.

b

$$\begin{aligned} E^s(y) &= - \int_{-1}^1 y(x) dx - \int_{-1}^1 p_3(x) dx \\ &= - \int_{-1}^1 \frac{y^{(4)}(\xi)}{4!} (x+1)x^2(x-1) dx \\ &= - \frac{y^{(4)}(\xi)}{90} \end{aligned}$$

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c

$$\int_a^b y(x) dx \approx \frac{h}{3} [y(a) + y(b) + 4 \sum_{i=1}^n y(x_{2i-1}) + 2 \sum_{i=1}^{n-1} y(x_{2i})]$$

$$E^s(y) = -y^{(4)}(\xi) \frac{(b-a)}{2880} (2h)^4 = -\frac{b-a}{180} h^4 y^{(4)}(\eta)$$

II

$$|f''(x) = e^{-x^2}(4x^2 - 2)| \leq 2, |f^{(4)}(x) = 4e^{-x^2}(4x^4 - 12x^2 + 3)| \leq 12.$$

a

$$\int_{-1}^1 f(x) dx = \frac{1}{2} \sum_{k=0}^{n-1} [f(x_k) + f(x_{k+1})] - \frac{b-a}{12} h^2 f''(\eta)$$

We already know that $f''(x) = e^{-x^2}(4x^2 - 2)$, so $|f''(\eta)| \leq 2$.

$$\frac{h^2}{6} < 0.5 * 10^{-6}$$

$$n^2 > \frac{1}{3} * 10^6$$

$$n > 577$$

b

From I, we have Simpson rule. We already know that $|f^{(4)}(x) = 4e^{-x^2}(4x^4 - 12x^2 + 3)| \leq 12$

$$\frac{h^4}{180} 12 < 0.5 * 10^{-6}$$

$$n^4 > \frac{2}{15} * 10^6$$

$$n > 19$$

III

a

$\pi_2(t)$ is orthogonal to $\pi_1(t) \in \mathbb{P}^1$.

$$\int_0^\infty (c_0 + c_1 t)(t^2 + at + b)e^{-t} dt = \int_0^\infty [(c_1 t^3 +) + (c_0 + c_1 a)t^2 + (ac_0 + c_1 b)t + c_0 b]e^{-t} dt = 6c_1 + 2c_1 a + 2c_0 + ac_0 + c_1 b -$$

$$\begin{cases} 2 + a + b = 0 \\ 6 + 2a + b = 0 \end{cases}$$

So, $a = -4, b = 2$, which means $\pi_2(t) = t^2 - 4t + 2$.

b

$\pi_2(t) = t^2 - 4t + 2 = 0$, we get $t_1 = 2 - \sqrt{2}, t_2 = 2 + \sqrt{2}$.

$$\begin{cases} \int_0^\infty e^{-x} dx = 1 = A_1 + A_2 \\ \int_0^\infty te^{-x} dx = 1 = A_1(2 - \sqrt{2}) + A_2(2 + \sqrt{2}) \end{cases}$$

So, $A_1 = \frac{2+\sqrt{2}}{4}, A_2 = \frac{2-\sqrt{2}}{4}$ and $E_2(f) = \frac{f^{(4)}(\xi)}{4!} \int_0^\infty \pi_2^2(x) e^{-x} dx = \frac{f^{(4)}(\xi)}{6}$.

c

We have $f(x) = \frac{1}{1+x}, f(x_1) = \frac{3+\sqrt{2}}{7}, f(x_2) = \frac{3-\sqrt{2}}{7}$.

$$\int_0^\infty \frac{1}{1+x} e^{-x} dx = \frac{3+\sqrt{2}}{7} A_1 + \frac{3-\sqrt{2}}{7} A_2 = \frac{4}{7}$$

$E_2(f) = I - \frac{4}{7}$, so we can get $24(1+\tau)^{-5} = 6(I - \frac{4}{7}), \tau = 1.0926$

IV

a

The basic Lagrange polynomial $\ell_m(t)$ is defined as:

$$\ell_m(t) = \prod_{\substack{j=1 \\ j \neq m}}^n \frac{t - x_j}{x_m - x_j}.$$

According to the problem, $h_m(t)$ and $q_m(t)$ are of the form:

$$h_m(t) = (a_m + b_m t) \ell_m^2(t), \quad q_m(t) = (c_m + d_m t) \ell_m^2(t).$$

To satisfy the Hermite interpolation conditions $p(x_m) = f_m$ and $p'(x_m) = f'_m$, we require:

$$h_m(x_m) = 1, \quad h'_m(x_m) = 0.$$

Substituting $h_m(t) = (a_m + b_m t) \ell_m^2(t)$, we obtain:

$$h_m(x_m) = a_m + b_m x_m = 1,$$

$$h'_m(x_m) = (b_m + 2a_m \ell'_m(x_m)) \ell_m^2(x_m) = b_m + 2a_m \ell'_m(x_m) = 0.$$

Since $\ell'_m(x_m) = \frac{d}{dt} \ell_m(t) \Big|_{t=x_m}$ and $\ell_m(x_m) = 1$, we have:

$$b_m = -2a_m \ell'_m(x_m) = -2\ell'_m(x_m).$$

Thus, the constants for $h_m(t)$ are:

$$a_m = 1 + 2x_m \ell'_m(x_m), \quad b_m = -2\ell'_m(x_m).$$

Similarly, to satisfy the Hermite interpolation conditions, we require:

$$q_m(x_m) = 0, \quad q'_m(x_m) = 1.$$

Substituting $q_m(t) = (c_m + d_m t)\ell_m^2(t)$, we obtain:

$$\begin{aligned} q_m(x_m) &= c_m + d_m x_m = 0, \\ q'_m(x_m) &= (d_m + 2c_m \ell'_m(x_m))\ell_m^2(x_m) = d_m + 2c_m \ell'_m(x_m) = d_m = 1. \end{aligned}$$

Thus, the constants for $q_m(t)$ are:

$$c_m = -x_m, \quad d_m = 1.$$

[1]

b

$$\begin{cases} \omega_k = \int_a^b (a_k + b_k x) l_k^2(x) \rho(x) dx \\ \mu_k = \int_a^b (c_k + d_k x) l_k^2(x) \rho(x) dx \end{cases}$$

c

$$\mu_k = \int_a^b (c_k + d_k x) l_k^2(x) \rho(x) dx = \int_a^b (x - x_k) l_k^2(x) \rho(x) dx = 0$$

To ensure that $\mu_k = 0$ for each $k = 1, 2, \dots, n$, the following conditions must be imposed on the node polynomial or the nodes x_k :

1. **Symmetry Condition:** If the nodes x_k are symmetrically distributed around a point (e.g., $x_k = -x_{n-k+1}$), then the integral of $q_k(t)$ will be zero. This is because the term $q_k(t) = (c_k + d_k t)\ell_k^2(t)$ will integrate to zero due to symmetry.

2. **Orthogonality Condition:** The node polynomial $\pi(t)$ must be orthogonal to the polynomial $t\ell_k^2(t)$ with respect to the weight function $\rho(t)$. This ensures that the integral $\int_a^b t\ell_k^2(t)\rho(t) dt = 0$, which implies $\mu_k = 0$.

3. **Specific Node Distribution:** If the nodes x_k are chosen as the roots of a specific orthogonal polynomial (e.g., Legendre or Laguerre polynomials), then $\mu_k = 0$ is automatically satisfied due to the properties of these polynomials.

Acknowledgement

References

- [1] DeepSeek AI. *DeepSeek*. <https://www.deepseek.com/>. Accessed: 2024-12-18.